



**OLLSCOIL NA GAILLIMHE**  

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**MSc Thesis Title**

**By**

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## **Declaration**

I declare that the work presented in this thesis has not been submitted for any Degree or Diploma at this, or any other University and that the work described.

I confirm that all sources of information used in the preparation of this dissertation have been appropriately acknowledged and referenced in accordance with the required academic standards. Any assistance received during the completion of this work has been fully acknowledged.

I further declare that informed consent was obtained from the participant and/or her legal guardian prior to the commencement of the case study. All data were collected and handled in accordance with ethical guidelines, maintaining participant confidentiality and anonymity throughout the study.

Signed:..... Date:.....

## **Acknowledgements**

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I would also like to acknowledge the invaluable learning experiences gained throughout my placement, which played a significant role in the development of both my professional and academic skills.

Working closely with individuals with complex health and developmental needs provided me with a deeper understanding of the challenges faced by people with intellectual disabilities and neurological conditions such as epilepsy.

This experience enhanced my appreciation of the importance of individualized exercise prescription, patient-centred care, and multidisciplinary collaboration in achieving positive health outcomes.

Observing and participating in the assessment, planning, implementation, and monitoring of exercise interventions allowed me to apply theoretical knowledge acquired during the MSc programme in a practical setting.

## **Abstract**

Individuals with intellectual disability often experience reduced physical activity levels, lower functional capacity, and an increased prevalence of secondary health conditions. The presence of epilepsy may further complicate participation in exercise programmes due to concerns regarding safety, supervision, and seizure management. The purpose of this case study was to evaluate the implementation of an individualized exercise programme for a female participant with intellectual disability and mild epilepsy within a community-based setting.

A comprehensive assessment was conducted to evaluate the participant's functional capacity, mobility, balance, muscular strength, and ability to perform activities of daily living. Based on the findings, an individualized exercise intervention was developed, focusing on improving physical fitness, functional independence, and overall quality of life while ensuring appropriate safety measures related to epilepsy management.

The intervention consisted of supervised aerobic, resistance, balance, and flexibility exercises performed over the placement period. Progress was monitored through observational assessments and repeat functional testing. Improvements were observed in functional mobility, exercise tolerance, balance, and confidence during physical activity participation. The participant demonstrated good adherence to the exercise programme and tolerated the intervention without adverse events.

The findings of this case study suggest that appropriately prescribed and supervised exercise can be a safe and effective intervention for individuals with intellectual disability and mild epilepsy. Exercise programmes tailored to individual abilities and health needs may contribute to improved physical function, independence, and overall well-being. Further research involving larger populations is required to establish evidence-based exercise guidelines for this population.

In addition to the physical outcomes, the intervention aimed to encourage regular participation in structured exercise and promote positive attitudes towards physical activity. Individuals with intellectual disability frequently encounter barriers to exercise participation, including limited access to suitable programmes, communication difficulties, and reduced confidence in performing physical tasks.

Therefore, the exercise programme was designed to be enjoyable, achievable, and adapted to the participant's cognitive and physical abilities. Particular consideration was given to seizure risk management, exercise intensity monitoring, and environmental safety throughout all sessions.

Regular supervision and individualized progression ensured that the participant could engage safely while maximizing the potential benefits of exercise training.

Furthermore, the intervention emphasized the development of functional skills that could enhance independence in everyday activities and support long-term health maintenance

## List of commonly used abbreviations

| Abbreviation | Meaning                                               |
|--------------|-------------------------------------------------------|
| ACSM         | American College of Sports Medicine                   |
| BMI          | Body Mass Index                                       |
| ID           | Intellectual Disability                               |
| ICF          | International Classification of Functioning           |
| ROM          | Range of Motion                                       |
| 6MWT         | Six-Minute Walk Test                                  |
| WHO          | World Health Organization                             |
| DSM-5-TR     | Diagnostic and Statistical Manual of Mental Disorders |
| FITT         | Frequency, Intensity, Time and Type                   |
| ADL          | Activities of Daily Living                            |

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# **Chapter 1: Introduction and Literature Review**

## **1.1 Introduction**

Intellectual disability (ID) is a neurodevelopmental disorder marked by significant limitations in intellectual functioning and adaptive behavior, starting during the developmental period before the age of 18 (American Psychiatric Association, 2022). These limitations impact the conceptual, social, and practical skills needed for independent living and participation in daily activities. The prevalence of intellectual disability is estimated to be around 1 to 3% of the global population, making it one of the most common developmental disabilities worldwide (Maulik et al., 2011). People with intellectual disability often have lower levels of physical activity, impaired motor skills, reduced cardiovascular fitness, decreased muscle strength, and a higher risk of other health problems compared to the general population (Hilgenkamp et al., 2012).

Epilepsy is a common neurological condition that occurs alongside intellectual disability. The prevalence of epilepsy in individuals with intellectual disability is significantly higher than in the general population, with studies showing rates between 20% and 30%, depending on the severity of the disability and other neurological issues (Robertson et al., 2015). Epilepsy is a chronic neurological disorder marked by recurrent, unprovoked seizures due to abnormal and excessive electrical activity in the brain (Fisher et al., 2017). Tonic-clonic epilepsy, one of the most well-known types of seizures, has an initial tonic phase with muscle stiffening, followed by a clonic phase with rhythmic muscle contractions and relaxation. While mild tonic-clonic epilepsy can often be managed with antiepileptic medications, worries about seizures can lead to lower participation in physical activities and exercise programs (Arida et al., 2010).

Regular physical activity is increasingly seen as a crucial part of health management for individuals with intellectual disability and epilepsy. Exercise is linked to improvements in cardiovascular fitness, muscle strength, balance, functional mobility, psychological well-being, and overall quality of life (Shields et al., 2008). Furthermore, research indicates that properly prescribed exercise does not increase the frequency of seizures and may lead to better neurological and physical health outcomes for individuals with epilepsy (Arida et al., 2013).

The participant in this case study is a female adult diagnosed with intellectual disability from birth and mild tonic-clonic epilepsy. She receives ongoing support, including manual therapy and exercise rehabilitation. A significant part of her intervention is the use of the Motomed movement therapy device, which helps with active and passive movement of both the upper and lower limbs. Motomed-assisted exercise is increasingly used in neurological and rehabilitation settings to promote mobility, improve circulation, maintain joint range of motion, and improve muscle function in those with limited physical abilities (Werner et al., 2012).

Although there is growing evidence supporting exercise interventions in neurological rehabilitation, limited research has specifically looked at the combination of Motomed-assisted exercise and therapeutic methods for individuals with both intellectual disability and epilepsy. This case study aims to assess the effectiveness and practical application of a structured exercise program that includes Motomed training and manual therapy in enhancing functional capacity, mobility, and overall well-being in an adult female with intellectual disability and mild tonic-clonic epilepsy.

A case study approach was chosen because it allows for a detailed examination of an individual participant in a real clinical setting. This method gives valuable insights into intervention strategies, participant responses, and practical issues that larger quantitative studies may overlook. The findings may add to the growing evidence supporting personalized exercise plans for people with intellectual disability and epilepsy.

The dissertation is organized into several chapters. After this introduction, a thorough literature review explores intellectual disability, epilepsy, exercise participation, rehabilitation strategies, and the use of Motomed-assisted therapy. The methodology chapter describes the participant assessment processes and intervention design. The case study analysis details the participant's outcomes and progress during the intervention period. Finally, the discussion and conclusion interpret the findings in the context of existing literature and offer recommendations for clinical practice and future research.

## 1.2 Literature Review

### Intellectual Disability

#### Definition and Classification of Intellectual Disability

Intellectual disability (ID) is a lifelong neurodevelopmental condition characterized by significant limitations in intellectual functioning and adaptive behaviour, with onset occurring during the developmental period before the age of 18 years (American Psychiatric Association, 2022). Intellectual functioning refers to general mental abilities such as reasoning, learning, problem-solving, planning, abstract thinking, judgement, and academic learning. Adaptive behaviour encompasses conceptual, social, and practical skills required for independent functioning in everyday life (Schalock et al., 2021).

Historically, intellectual disability was classified primarily according to intelligence quotient (IQ) scores. However, contemporary diagnostic frameworks emphasize adaptive functioning as the most important determinant of severity and support needs (American Psychiatric Association, 2022). Intellectual disability is generally categorized into mild, moderate, severe, and profound levels based on the individual's functional abilities and degree of support required in daily activities (Schalock et al., 2021).

Individuals with mild intellectual disability typically achieve some degree of independence but may require assistance with complex decision-making, education, employment, and social interactions. Those with more severe forms often require extensive lifelong support. The participant included in this case study presents with intellectual disability from birth and demonstrates functional limitations that influence physical activity participation and overall independence.

### 1.2.1 Epidemiology

Intellectual disability affects approximately 1–3% of the global population, making it one of the most prevalent developmental disabilities worldwide (Maulik et al., 2011). The prevalence varies according to diagnostic criteria, geographical region, socioeconomic factors, and healthcare accessibility. Mild intellectual disability accounts for approximately 75–85% of all diagnosed cases (McKenzie et al., 2016).

Advances in healthcare and social support services have contributed to increased life expectancy among individuals with intellectual disability. Consequently, there is growing emphasis on promoting healthy ageing, maintaining functional independence, and reducing secondary health complications through targeted health interventions and exercise programmes (Heller et al., 2011).

### 1.2.2 Pathophysiology of Intellectual Disability

The pathophysiology of intellectual disability is complex and multifactorial. Intellectual disability results from disruptions in normal brain development occurring during prenatal, perinatal, or postnatal periods. These disruptions may affect neuronal proliferation, migration, differentiation, synaptic development, and neural connectivity, ultimately leading to impaired cognitive and adaptive functioning (Shevell et al., 2003).

Genetic factors account for a significant proportion of cases. Chromosomal abnormalities such as Down syndrome, Fragile X syndrome, Rett syndrome, and other genetic mutations have been strongly associated with intellectual disability (Ropers, 2010). Environmental factors may also contribute, including maternal infections during pregnancy, exposure to toxins, nutritional deficiencies, birth complications resulting in hypoxic brain injury, traumatic brain injury, and central nervous system infections (Shevell et al., 2003).

Neuroimaging studies have demonstrated structural and functional abnormalities in several brain regions among individuals with intellectual disability. These include alterations in cortical thickness, reduced brain volume, abnormalities in white matter pathways, and impaired connectivity between regions involved in learning, memory, executive functioning, and motor

control (Pavone et al., 2014). Such neurological impairments contribute to the cognitive, behavioural, and motor difficulties frequently observed in this population.

## Physical and Functional Characteristics

Individuals with intellectual disability commonly demonstrate reduced physical fitness compared with the general population. Studies have reported lower levels of cardiovascular endurance, muscular strength, flexibility, balance, coordination, and motor proficiency (Hilgenkamp et al., 2012). These deficits may be attributed to neurological impairments, sedentary lifestyles, reduced participation in physical activity, social barriers, and limited access to exercise opportunities.

Motor impairments are particularly relevant in rehabilitation settings. Difficulties with balance, postural control, gait, coordination, and movement planning can restrict participation in activities of daily living and reduce independence (Hartman et al., 2010). Furthermore, individuals with intellectual disability often exhibit lower levels of physical activity and higher rates of obesity, cardiovascular disease, diabetes, and musculoskeletal disorders than their peers without disability (Temple et al., 2010).

### 1.2.3 Physical Activity and Exercise in Intellectual Disability

Regular physical activity is recognized as an important component of health promotion and rehabilitation for individuals with intellectual disability. Exercise interventions have demonstrated significant benefits in cardiovascular fitness, muscular strength, balance, mobility, body composition, mental health, and quality of life (Shields et al., 2008).

Aerobic exercise programmes have been shown to improve cardiovascular function and endurance, while resistance training can increase muscular strength and functional capacity (Calders et al., 2011). Balance and coordination exercises may reduce fall risk and enhance mobility, contributing to greater independence in daily activities (Hartman et al., 2010).

Despite these benefits, participation rates remain low. Barriers include cognitive limitations, lack of accessible facilities, transportation difficulties, inadequate programme availability, insufficient caregiver support, and concerns regarding safety (Temple et al., 2010). Consequently, individualized exercise programmes tailored to the participant's abilities and support needs are recommended as best practice.

### Current Best Practice for Intellectual Disability

Current international guidelines advocate regular participation in physical activity for individuals with intellectual disability. The World Health Organization recommends at least 150 minutes of moderate-intensity physical activity per week for adults with disabilities, alongside muscle-strengthening activities performed on two or more days per week (World Health Organization, 2022).

Exercise prescription should be individualized, person-centred, and adapted according to functional ability, health status, and personal goals. Programmes should emphasize enjoyment, safety, gradual progression, and long-term adherence. Supervision by appropriately trained professionals may be required to optimize participation and minimize risks (Shields et al., 2008).

For individuals presenting with co-existing neurological conditions such as epilepsy, additional precautions may be necessary to ensure safe exercise participation while maximizing therapeutic benefits. Therefore, understanding the interaction between intellectual disability and epilepsy is essential when designing rehabilitation programmes for this population



## 1.3 Epilepsy

### 1.3.1 Definition and Classification

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures resulting from abnormal electrical activity within the brain (Fisher et al., 2017). It affects approximately 50 million people worldwide and represents one of the most common neurological disorders globally (World Health Organization, 2023).

Seizures are classified according to their site of origin within the brain and their clinical presentation. They may be focal, generalized, or of unknown onset. Generalized tonic-clonic seizures involve widespread neuronal activation affecting both cerebral hemispheres and are characterized by loss of consciousness, muscle stiffening, and rhythmic muscle contractions (Fisher et al., 2017).

The participant in this case study has a diagnosis of mild tonic-clonic epilepsy, which is managed through ongoing medical supervision and supportive care.

### 1.3.2 Pathophysiology of Epilepsy

The underlying pathophysiology of epilepsy involves an imbalance between excitatory and inhibitory neural activity within the central nervous system. Normal brain function depends on precise regulation of neuronal excitation and inhibition. In epilepsy, excessive excitatory activity or reduced inhibitory control results in abnormal synchronized neuronal discharges that manifest clinically as seizures (Stafstrom and Carmant, 2015).

Several mechanisms may contribute to epileptogenesis, including alterations in ion channel function, neurotransmitter abnormalities, structural brain lesions, genetic mutations, inflammatory processes, and changes in synaptic plasticity (Stafstrom and Carmant, 2015). The neurotransmitter glutamate plays a major role in neuronal excitation, whereas gamma-aminobutyric acid (GABA) serves as the primary inhibitory neurotransmitter. Dysregulation of these systems can increase neuronal excitability and seizure susceptibility.

During a tonic-clonic seizure, widespread cortical activation initially produces the tonic phase, characterized by sustained muscle contraction and rigidity. This is followed by the clonic phase, involving repetitive cycles of muscle contraction and relaxation. Postictal symptoms

such as fatigue, confusion, and muscle soreness may occur following seizure activity (Fisher et al., 2017).

## Epilepsy and Intellectual Disability

The relationship between intellectual disability and epilepsy is well established. Epilepsy occurs at significantly higher rates among individuals with intellectual disability than in the general population (Robertson et al., 2015). Shared underlying neurological abnormalities, developmental brain disorders, genetic syndromes, and structural brain injuries may contribute to the coexistence of both conditions.

The presence of epilepsy can further impact functional ability, physical activity participation, cognitive performance, and quality of life. Consequently, rehabilitation programmes for individuals with intellectual disability must consider both conditions simultaneously to optimize safety and effectiveness

## 1.4 Exercise in Intellectual Disability and Epilepsy

### Historical Perspectives on Exercise Participation

Historically, individuals with epilepsy were often discouraged from participating in physical activity because of concerns regarding seizure occurrence, injury risk, and sudden unexpected death in epilepsy (SUDEP). For many years, healthcare professionals and caregivers believed that exercise could trigger seizures and potentially worsen neurological symptoms. Consequently, many individuals with epilepsy adopted sedentary lifestyles, contributing to reduced physical fitness and increased health complications (Arida et al., 2010).

However, research conducted over the past three decades has challenged these assumptions. Current evidence suggests that appropriately prescribed and supervised exercise is generally safe for most individuals with epilepsy and may provide substantial physical, psychological, and neurological benefits (Capovilla et al., 2016). Modern clinical guidelines now encourage regular physical activity participation while considering individual seizure characteristics and medical status.

## Physiological Benefits of Exercise in Epilepsy

Regular exercise contributes to improvements in cardiovascular health, muscular strength, endurance, flexibility, balance, and overall physical functioning. These benefits are particularly important for individuals with epilepsy, who often experience lower physical fitness levels than the general population (Arida et al., 2013).

Aerobic exercise has been shown to improve cardiorespiratory fitness, increase oxygen utilization, and enhance functional capacity. Resistance training contributes to improved muscle strength, joint stability, and mobility, while balance and coordination exercises may reduce fall risk and improve confidence during movement (Nakken, 2008).

In addition to physical adaptations, exercise is associated with positive neurological effects. Studies suggest that regular physical activity may influence neurotransmitter regulation, increase cerebral blood flow, promote neuroplasticity, and enhance cognitive functioning (Arida et al., 2013). Exercise may also reduce stress and anxiety, which are recognized seizure triggers in some individuals.

## Exercise and Seizure Frequency

One of the primary concerns regarding exercise participation in epilepsy is the possibility of seizure provocation. However, evidence indicates that exercise rarely triggers seizures and may, in some cases, contribute to reduced seizure frequency (Arida et al., 2010).

Several mechanisms have been proposed to explain this protective effect. Exercise may increase inhibitory neurotransmitter activity, improve autonomic nervous system regulation, reduce psychological stress, and enhance overall neurological stability. Furthermore, improved sleep quality resulting from regular physical activity may indirectly contribute to better seizure control (Arida et al., 2013).

Although exercise is generally considered safe, individualized risk assessment remains essential. Factors such as seizure type, seizure frequency, medication use, and environmental conditions should be considered when designing exercise programmes.

#### 1.4.1 Exercise Prescription Considerations (ACSM)

Exercise programmes for individuals with epilepsy should follow established exercise physiology principles while incorporating condition-specific safety precautions. Programmes should begin at appropriate intensity levels and progress gradually according to individual tolerance and functional capacity.

Safety considerations include:

- Appropriate supervision during exercise sessions.
- Monitoring for signs of fatigue and overexertion.
- Ensuring adequate hydration.
- Avoiding exercise environments that may increase injury risk.
- Maintaining adherence to prescribed antiepileptic medication.
- Educating caregivers and support staff regarding seizure first aid.

Current recommendations suggest that most individuals with well-controlled epilepsy can safely participate in aerobic exercise, resistance training, flexibility exercises, and recreational physical activity (Capovilla et al., 2016).

For the participant in this case study, exercise interventions were implemented within a controlled rehabilitation environment with continuous supervision and appropriate monitoring. This approach minimized potential risks while facilitating active participation in therapeutic exercise.

## 1.5 Motomed Therapy and Neurological Rehabilitation

### Overview of Motomed Therapy

Motomed is a motor-assisted movement therapy device designed to facilitate repetitive, cyclical movement of the upper and lower extremities. The system allows individuals with neurological, musculoskeletal, or functional impairments to engage in active, active-assisted, or passive exercise depending on their abilities and rehabilitation goals (Werner et al., 2012).

The device consists of a motor-driven ergometer that can be used for arm or leg cycling. Unlike traditional exercise equipment, Motomed systems can provide assistance when voluntary movement is limited while simultaneously encouraging active participation whenever possible. This makes the technology particularly suitable for individuals with neurological disorders, intellectual disabilities, reduced mobility, and chronic health conditions.

In the present case study, Motomed exercise represented the primary intervention strategy and was utilized to promote movement of both the upper and lower limbs.

### Physiological Principles of Motomed Training

The therapeutic effects of Motomed training are based on the principle of repetitive task-specific movement. Repetitive movement stimulates neuromuscular activation, enhances circulation, maintains joint mobility, and promotes muscular endurance.

Passive cycling may assist individuals who have limited voluntary movement by providing continuous limb motion. Active-assisted modes encourage participation while providing support when fatigue or weakness occurs. Active modes require the participant to generate movement independently and may contribute to improvements in muscular strength and cardiovascular fitness (Werner et al., 2012).

Regular participation in cycling-based rehabilitation can increase blood flow to working muscles, improve oxygen delivery, maintain range of motion, and reduce the adverse effects of prolonged inactivity.

## Upper Limb Benefits of Motomed Therapy

Upper limb dysfunction is common among individuals with neurological and developmental disabilities. Reduced coordination, muscle weakness, impaired motor control, and limited movement opportunities can negatively affect activities of daily living and independence.

Motomed upper limb training provides repetitive arm cycling movements that encourage bilateral coordination and continuous upper extremity activation. Potential benefits include:

- Improved shoulder and elbow mobility.
- Enhanced muscular endurance.
- Increased circulation.
- Improved coordination and motor control.
- Maintenance of functional upper limb movement.

Regular upper limb training may support functional activities such as reaching, lifting, carrying objects, and performing self-care tasks.

## Lower Limb Benefits of Motomed Therapy

Lower limb training represents one of the most common applications of Motomed therapy. Repetitive cycling movements activate major muscle groups of the hips, thighs, and lower legs while minimizing impact forces on joints.

Research suggests that lower limb cycling interventions may improve:

- Functional mobility.
- Gait performance.
- Lower limb strength.
- Cardiovascular endurance.
- Balance and postural control.
- Circulatory function.

For individuals with intellectual disability who may have reduced participation in conventional exercise programmes, Motomed provides a structured and accessible method of achieving regular physical activity.

### Motomed Therapy in Neurological Conditions

The effectiveness of Motomed-assisted exercise has been investigated in several neurological populations, including stroke, cerebral palsy, Parkinson's disease, multiple sclerosis, and acquired brain injury. Findings generally demonstrate improvements in mobility, muscle activation, functional capacity, and exercise tolerance (Bressel et al., 2018).

Although research specifically involving individuals with intellectual disability and epilepsy remains limited, the physiological principles underlying Motomed therapy suggest potential benefits for this population. The controlled nature of the intervention allows exercise participation while reducing injury risk and facilitating individualized progression.

For individuals with intellectual disability, the repetitive and predictable movement patterns provided by Motomed may improve engagement and adherence compared with more complex exercise modalities. Furthermore, the ability to perform exercise in seated and supervised conditions makes the device particularly suitable for individuals requiring additional support during rehabilitation



## **Chapter 2: Methodology**

### **2.1 Research Design**

For this study, a single-case study design was used to examine the efficacy of an individualized exercise intervention for a female adult who has been diagnosed with intellectual disability and mild tonic-clonic epilepsy. A case study research design was chosen because it allows the researcher to examine an individual participant in depth within the context of a realistic clinical environment by taking into account the complexity of the interplay between several health conditions. While experimental designs tend to lack detail about assessment, intervention, clinical reasoning, and participant progress, case studies have much greater value in rehabilitation and exercise physiology contexts (Yin, 2018).

Since people with intellectual disability possess varied physical, cognitive, and behavioral traits, an individualized approach to assessment and prescription of exercises is necessary. Epilepsy adds even more complexity to the rehabilitation process due to the need to manage seizures, take into account fatigue, and ensure safety during the entire process of intervention. Therefore, an individualized case study approach is suitable for such evaluations (Thomas, 2021).

The first objective of this case study was to measure the effectiveness of an exercise physiology program using Motomed-assisted exercises on the upper and lower limbs in an individual with intellectual disability and mild tonic-clonic seizures who is being attended at a community rehabilitation service. The other objectives were the measurement of the ability to walk functionally, grip strength in the upper limbs and exercise tolerance over an eight-week period.

In this case study, there was both a quantitative aspect and the clinical observation aspect involved. In the quantitative evaluation, Six Minute Walk Test (6MWT) and hand grip strength test were used. Clinical observation of exercise tolerance, participant participation, and functional mobility were taken during the eight weeks of intervention

## 2.2 Placement Setting

This placement was carried out in Ability West, Mountbellew, County Galway, Ireland. Ability West is a community-based organization that offers person-centered services for children and adults who suffer from intellectual disability and autism. The organization seeks to ensure independence, inclusion and quality of life through rehabilitation, healthcare, educational and community support. This placement was carried out during eight weeks in June and July 2026. It was carried out on Tuesdays and Thursdays from 09:00 to 17:30 with supervision by qualified healthcare professionals. Through this placement the student was able to observe multidisciplinary rehabilitation programs as well as participate in exercise interventions under supervision.

Services offered in exercise physiology included ensuring functional independence, mobility, muscular endurance and physical activity participation. Exercise programs were tailored to suit the medical history, physical capabilities and goals of the individual. Special attention was paid to ensuring that people with complex disabilities are involved in safe and fun exercises.

The individual that was included in this case study adhered to the exercise schedule that was given and participated well in the intervention programme.

## 2.3 Participant Selection

Purposive sampling was employed in selecting the participant for the case study. Purposive sampling is typically employed in the case studies conducted in clinical settings as it allows for choosing a participant that would be capable of providing adequate information about the particular intervention (Yin, 2018).

The participant was a 25-year-old female with a diagnosis of intellectual disability and mild tonic-clonic epilepsy. She visited Ability West on a regular basis to receive treatment and social assistance and was determined by the multidisciplinary team to be medically fit to participate in the exercise intervention under supervision.

The baseline anthropometric characteristics were measured prior to the beginning of the intervention. The participant's height was 155 cm, with a body mass of 70 kg, resulting in the body mass index of 29.1 kg/m<sup>2</sup> (the range of BMI indicating that the participant is overweight as per WHO classification) (World Health Organization, 2022).

### Inclusion Criteria

The participant fulfilled the following inclusion criteria:

- Diagnosis of intellectual disability.
- Diagnosis of medically managed mild tonic-clonic epilepsy.
- Medically stable with no recent seizure-related complications.
- Regular attendance at Ability West.
- Ability to follow simple verbal instructions.
- Capacity to participate safely in supervised exercise sessions.
- Informed consent obtained from the participant's legal guardian.

### Exclusion Criteria

Participants would normally be excluded if they presented with:

- Uncontrolled epilepsy with frequent seizures.
- Acute musculoskeletal injury preventing exercise.
- Unstable cardiovascular disease.
- Acute illness or infection.
- Medical advice restricting participation in exercise.

No exclusion criteria applied to the participant selected for this study

## 2.4 Ethical considerations

The case study followed all ethical procedures at all levels of this study. The research has adhered to the ethical procedures that have been set out within the Declaration of Helsinki on research on human participants (World Medical Association, 2013).

Before conducting the study, clearance was obtained from the placement organisation and the exercise physiologist supervising the student on their placement. Both the participant's legal guardian was given written and verbal explanations about the purpose of the case study and the exercise program, and how the collected data would be used.

Confidentiality of the participant has been kept intact in this dissertation. No personal details such as names, addresses or dates of birth have been given. All of the clinical data provided has been anonymous, and the participant has only been identified through their demographics. Data collected from the placement was stored safely and available to only the student and supervising personnel.

Participation in the exercise program was purely optional. The participant had the liberty to discontinue participation in both the intervention and the case study at any point without compromising his ability to receive rehabilitation services through normal channels. During the intervention, the safety of the participant was the topmost priority.

The exercise program was closely monitored, and should signs of exhaustion, pain or seizures be witnessed, the exercise program would be stopped immediately.

The case study adhered to the principles of beneficence, non-maleficence, autonomy and justice, ensuring that the participant's wellbeing remained central throughout the rehabilitation programme.

## 2.5 Baseline Assessment and Outcome Measures

Before conducting the exercise programme, a complete baseline assessment was performed in order to determine the physical condition of the participant and develop a tailored exercise programme. The baseline assessment is an important element of exercise programming that gives valuable objective information about the level of functional

fitness, muscle strength, exercise tolerance and physical limitations, which can help professionals to track the progress during the rehabilitation process (American College of Sports Medicine (ACSM), 2022).

The participant was a 25-year-old woman, having a height of 155 cm and a body weight of 70 kg. Consequently, her body mass index was calculated as 29.1 kg/m<sup>2</sup>. Anthropometric measures were obtained by standard methods. Body mass index is an easy way to assess body weight and detect people who might have the high risk of diseases connected with obesity (World Health Organization, 2022). Even though BMI does not reflect the body composition of an individual, it still remains an informative tool in rehabilitation practice.

Having obtained anthropometric data, a sub-maximal exercise test, namely the Six-Minute Walk Test (6MWT) was applied. The 6MWT is a proven submaximal exercise test commonly used to assess functional exercise capacity in different types of neurological, cardiac, and respiratory diseases (American Thoracic Society, 2002). While performing the test, the participant had to cover a fixed route in a pre-established room within six minutes while walking at his/her chosen pace under close supervision of the examiner. Verbal encouragement was used during the test in accordance with the testing guidelines.

The Six-Minute Walk Test was chosen since it allows evaluating a patient's ability to conduct daily activities, not maximal physical efforts. According to previous research, the 6MWT has good reliability and validity and can be used as a sensitive tool in assessing patients' response to rehabilitation programs (ATS, 2002). The participant managed to pass the test successfully showing good results in walking performance. No complications or seizures happened during the test.

Function of upper limbs muscles was assessed through handgrip strength test. Handgrip strength has been widely accepted as an accurate tool for assessing muscular strength and functional capacity (Roberts et al., 2011). Lower grip strength in patients has been

correlated with decreased independent functional ability, poor functional outcomes and diminished quality of life in neurological conditions. Thus, the handgrip strength test was selected as a measure of upper limb function during the intervention period.

The grip strength of the patient was measured with the use of a hand dynamometer with calibrated settings. The patient was asked to sit comfortably with the shoulder adducted, elbow flexed at about 90 degrees and forearm in neutral position. Three maximum efforts were made and sufficient recovery periods were allowed between each effort. The greatest value from three measurements was recorded. This standardized test protocol increases the reliability of the measurement (Roberts et al., 2011).

A balance assessment was taken into consideration for the first assessment. However, as the participant had an intellectual disability and poor comprehension of instructions, along with safety concerns, balance tests such as the Berg Balance Scale test were ruled out from being conducted. Instead, balance and postural control were monitored by clinical observation of the participant during exercise participation.

Range of joint movement was assessed via observation during exercise participation. Major joints of both the upper and lower body were examined during active movements in order to detect any limitations which would affect exercise participation. No major changes in joint movement were seen during the period of the intervention program.

Tolerance to exercise was assessed continually for eight weeks through the observation of how well the participant performed the specified exercise regime without undue fatigue, pain, or signs of seizure. Observations made were willingness to exercise, the ability to sustain exercise duration, post-exercise recovery and participation in the exercise session.

Combining objective assessment with clinical observations offered an all-round assessment

of the individual's physical capability and thus helped the exercise programme be progressed based on their capacity and tolerance level.

## 2.6 Exercise Intervention

After completion of the baseline assessment, a personalized exercise program was formulated based on the physical ability, medical history and goals of the participant. The exercise prescription was done following the guidelines provided by the American College of Sports Medicine (ACSM, 2022). The Frequency, Intensity, Time and Type (FITT) principle was used to make sure that the intervention was safe and effective for the participant's functional ability.

This intervention was conducted for a duration of eight weeks during the student's placement at Ability West. Exercises were performed two times per week on Tuesdays and Thursdays under the guidance of healthcare professionals. Each exercise session took about 30-40 minutes, depending on the physical ability and motivation of the participant.

The main objective of this intervention was to enhance functional mobility, muscle endurance, upper limb strength and physical activity participation for the patient with mild tonic-clonic epilepsy.

Warm up was done in every session of exercise and consisted of gentle active exercise of upper and lower extremities for 5 minutes. Warm up exercises included shoulder flexion/extension, shoulder circles, elbow flexion/extension, wrist mobility exercises, seated marching, and ankle movement exercises. Warm up exercises were performed with low intensity in order to gradually increase blood flow and prepare the cardiovascular system for exercise and prevent musculoskeletal injuries (ACSM, 2022).

The main part of the intervention included Motomed-assisted exercises. The Motomed movement therapy machine was chosen due to its ability to provide repetitive active arm

cycling and leg cycling in a safe sitting position for individuals with neurological disabilities. Previous studies have shown that repetitive cycling exercises might help to develop muscular endurance, increase blood circulation, motor control and functional mobility in neurological rehabilitation (Werner et al., 2012).

Training of upper limbs was done before training of lower limbs in every session. Participant cycled actively with arms on Motomed machine for 14 minutes with resistance. Participants were advised to cycle steadily and were provided with verbal encouragement from a supervisor during exercising. Rest periods were allowed in case of fatigue.

After completing the upper limbs exercise program, the participant proceeded to do lower limb cycling on the Motomed for seven minutes at the same resistance levels. Lower limb cycling helped in repetitive use of muscles in the hips, knees, and ankles while ensuring coordination of bilateral movement. Cycling in a seated posture increased stability and reduced chances of falls during the exercise process.

Intensity of exercise was kept moderate and light throughout the intervention period. Due to the participant's intellectual disability, subjective measures such as Borg Rating of Perceived Exertion scales could not be reliable all through the process. Intensity of exercise was therefore monitored by observing facial expression, breathing rate, movements, and ability to tolerate exercises. Observational technique is usually employed in people with cognitive impairment and communication problems (ACSM, 2022).

Exercise progression was individually designed based on the ability of the participant to tolerate the exercise and regular attendance. Instead of increasing resistance significantly, progression involved encouraging better cycling skills and ensuring the participant maintains participation in the activity without taking many breaks. This was seen as the most appropriate way of progression for the participant due to the condition of his nervous system.

A cool-down process taking about five minutes was performed at the end of each session.



This process involved slow active limb movements, breathing exercises, and sitting relaxation in order to allow gradual lowering of the heart rate after exercise.

During the intervention the participant showed proper compliance with the exercise programme, and attended all exercise sessions. During these exercise sessions participants were observed by professionals who assisted the participant with exercise and watched him carefully for possible occurrence of seizures, fatigue and other adverse reactions to exercise. No adverse effects were observed in the course of the intervention period, and the participant performed all Motomed exercise sessions.

The designed exercise programme helped the participant to increase his physical activity participation in a proper way according to his needs related to the intellectual disability and epilepsy he had.

## 2.7 Safety of Exercises and Supervision

Exercise safety has been seen as an important factor during the intervention process since the participant has been diagnosed with tonic-clonic epilepsy of mild form and intellectual disability. Prior to conducting every exercise session, the general state of health of the participant, her desire to take part in exercises and her general physical state has been examined by the supervising staff members. Any signs of sickness, tiredness, dizziness or behavioral changes indicating that the participant cannot do exercises on a certain day have been noticed and taken into account. Individual exercise prescription for people suffering from epilepsy should be provided in a safe environment with supervision (Capovilla et al., 2016).

Constant supervision during all the exercise sessions was provided by the trained healthcare workers. The participant used the Motomed sitting down; thus, there has been no risk of falls and injuries. Sitting posture of the Motomed machine is especially helpful for people with neurological conditions because it gives trunk support and allows doing repetitive limb movements (Werner et al., 2012).

Intensity of exercises was measured using direct clinical assessment rather than setting heart rate limits or through the use of subjective measurement tools such as ratings of perceived exertion (RPE). Considering that the participant suffers from mental retardation and cognitive impairments, it seemed reasonable to evaluate exercise tolerance on the basis of facial expression, respiration, skin color, speech, body movements, and over-fatigue. Exercise was stopped once the participant started to feel tired or asked for a break. Attention was paid to managing seizures during exercise sessions. Healthcare professionals were aware of the medical background of the patient and procedures of seizure management that had been developed at Ability West. The environment where exercise took place was not cluttered and safe, with enough space provided around the Motomed equipment. Luckily, no seizures happened during any of the supervised exercise sessions.

Sufficient water intake was recommended before and after the exercise programme, and enough recovery time was allowed between exercises. The participant did very well on the exercise programme, and there were no complications.

## 2.8 Data Collection and Outcomes Assessment

Collection of data took place at baseline and then upon completion of the intervention programme which lasted for eight weeks. The quantitative measures used were anthropometric measures, Six-Minute Walk Test (6MWT), and hand grip strength measurement. The clinical observations on endurance, exercise tolerance, and functional mobility were taken after each exercise session under supervision.

The Six-Minute Walk Test was chosen as the main test for measuring functional walking capacity since it represents the patient's ability to complete everyday activities with a high level of reliability (American Thoracic Society, 2002). Upon completing the intervention programme, the participant showed an improvement in their walking performance compared to baseline testing. Even though the patient had a good baseline score on walking performance, there was still an improvement in walking endurance and exercise tolerance

Assessment of grip strength was also undertaken after the intervention, to assess any improvements in upper limb muscle function. An increase in grip strength indicated that there had been an improvement in muscle activation in the upper limb muscles, which could be attributed to active upper limb cycling in the Motomed exercise.

Exercise endurance was assessed clinically during the placement. At the beginning of the intervention, the client needed some rest during the exercise, but at the end of the intervention, she showed some improvement in the tolerance of continuous activity without rest periods. Willingness to exercise and consistency in exercising were also noted.

The joint range of motion did not change much over the intervention period, since the main objective of the programme was not the improvement of the joint range of motion. The objective was the improvement of functional activity and exercise tolerance.

Formal balance assessment was not undertaken owing to cognitive difficulties of the patient as well as safety issues. However, an observational assessment showed that the patient remained balanced during the exercises and displayed confidence when transferring to and from the Motomed machine.

## 2.9 Data Analysis

Considering the character of a single-participant case study, descriptive data analysis was used to determine the effectiveness of the intervention. Case study approach focuses on the interpretation of the individual participant's reaction rather than comparing groups statistically (Yin, 2018).

Changes in participants' performances at baseline and after intervention were described and interpreted in conjunction with their clinical observations during supervised activities. Participants' attendance, compliance, tolerance and progress through the course of the intervention period were also taken into account while assessing the program effectiveness.

Since there was only one participant in this study, inferential statistical analysis could not be used. Thus, the results were described and interpreted with the help of tables and descriptions of changes after completing the intervention program.

## 2.10 Reliability and Validity

There were several approaches used to improve the reliability and validity of the current case study. Firstly, all tests were done following standard procedure based on clinical guidelines available in literature. In particular, Six-Minute Walk Test was conducted following the American Thoracic Society (2002) guidelines while grip strength test was done by means of positioning and performing the test as per Roberts et al. (2011).

Secondly, all assessments were completed in the same rehabilitation environment under similar conditions using the same piece of equipment both before and after the intervention. This approach ensured that no variability in the measurements and improved reliability of collected data occurred.

Finally, the exercises were conducted by the same placement team during the eight-week long intervention period. In addition, resistance level of the Motomed and duration of the training session were kept constant unless changes were needed due to the fatigue or comfort of the participant.

Observations of the participant's clinical status were recorded throughout the intervention period to supplement the objective outcome measures. Combination of qualitative and quantitative measures enhanced the validity of the evaluation process as the participant progress could be better assessed.

Even though results from a single case study are not generalizable to the whole population, they contribute greatly to gaining an understanding of individualized exercise and rehabilitation for individuals with intellectual disabilities and epilepsy. This case study is relevant because it adds clinical information on the use of Motomed exercise within the community environment.

## 2.11 Chapter Summary

This chapter has elaborated on the methodology used in evaluating the effectiveness of the individualised exercise intervention for a 25-year-old woman with intellectual disability and mild tonic-clonic epilepsy who attends Ability West in Mountbellew. The single participant case study method has been chosen in order to enable thorough examination of the exercise prescription in a real life clinical setting. The baseline measurements comprised anthropometric assessment, the six-minute walk test and hand grip strength, whereas exercise tolerance and functional mobility have been assessed via continuous clinical observation.

The intervention entailed eight weeks' exercise programme involving upper and lower limb active Motomed training two times per week. Safety of the patient has been a top priority in every exercise session, as appropriate supervision and monitoring were undertaken to reduce the likelihood of any adverse incidents. Descriptive statistics have been employed in comparing the baseline and post-intervention measurements in order to obtain evidence-based evaluation of the functional status following rehabilitation. Chapter four provides the results of the intervention and examines changes in the participant's functional status.

## **Chapter 3: Results and Case Study Analysis**

### **3.1 Participant Description**

The participant was a 25-year-old woman attending Ability West, Mountbellew, County Galway, with a diagnosis of an intellectual disability since birth and mild tonic-clonic epilepsy under medical management. The intellectual disability was a challenge in the cognitive performance, language and ability to carry out some of her day to day activities independently. Nevertheless, the participant showed positivity towards physical exercise and enthusiasm while engaging in the exercises under supervision.

At the beginning of the intervention, the height of the participant was 155 cm with a body weight of 70 kg; hence her body mass index (BMI) was about 29.1 kg/m<sup>2</sup>. Despite falling within the overweight category as per the World Health Organization classification, the participant continued being physically active in her daily programme and healthy enough for participation in the exercise.

The participant attended Ability West twice per week as a part of the rehabilitation and community support programme for her condition. The exercise intervention was carried out as a part of the exercise physiology placement program, and the exercise interventions were conducted under the supervision of health care professionals for eight weeks. Due to certain unavoidable factors, the participant could not attend two of the exercise interventions; however, she participated in all other exercise interventions with high levels of adherence. The primary rehabilitation objectives for the participant were to ensure mobility, develop muscular endurance, increase involvement in physical activities and achieve functional independence. Since the epilepsy was well managed through medication, exercise intervention was considered safe.

### 3.2 Baseline Assessment Results

Prior to starting the program, baseline assessments had been done in order to determine the physical capabilities of the participant.

#### Anthropometric Assessment

The results from the anthropometric assessment revealed that the weight of the participant is 70 kg and her height is 155 cm. From the result of the calculation for body mass index, it can be inferred that weight control can be included as one of the considerations in health promotion. People with intellectual disability usually have higher prevalence of being overweight and obese.

#### Walking Functional Capacity

The functional exercise capacity was evaluated through the Six-Minute Walk Test (6MWT). The individual performed the test successfully and at a steady walking pace for six minutes without any problems. Her walking capacity is deemed to be normal according to her ability and health condition.

The individual had minimal need for verbal encouragement and motivation during the test. No balance problems, excessive fatigue and neurologic manifestations were observed. In addition, no seizure activity was observed either before, during or after the test. While baseline walking capacity was satisfactory, improvement could still be gained in terms of her endurance, exercise capacity and confidence when she participates regularly in exercise programs.

## Upper Limb Grip Strength

Strength of upper limbs was assessed via hand grip dynamometry. The results from grip strength test revealed adequate functioning of the muscles of the upper limbs in both hands. The participant showed understanding and cooperation while being able to perform maximum efforts in the course of testing. Despite the fact that the grip strength at the beginning was already within the range of acceptable functionality, further development of muscular endurance and neuromuscular coordination was possible due to repetitiveness of exercise with the Motomed machine.

## Balance and Functional Mobility

No formal assessment of balance was conducted as standardized measures of balance were deemed inappropriate for the participant based on their cognitive impairments and potential safety concerns. Functional mobility and postural control were therefore assessed by clinical observation.

It was observed that the participant could mobilize independently in the rehabilitation setting and exhibited stable sitting posture while exercising. Safe functional transfer on and off the Motomed machine was conducted.

## Range of Motion

Evaluation of the range of motion visually by observing movement of the upper and lower extremities confirmed that there were no restrictions in movements and joints functioned properly. Active movements of the shoulder joints, elbow joints, hip joints, knee joints and ankle joints were sufficient to engage in the exercise program.



## Clinical Presentation

Generally, the participant demonstrated willingness to be involved in physical exercises. She responded positively to positive reinforcement in terms of verbal feedback and enjoyed physical activities in the rehabilitative setting. Since she had her epilepsy under medical control without any recorded incidences of seizures at Ability West, the participant was deemed fit to undergo supervised exercises.

### 3.3 Description of the Exercise Intervention Programme

The exercise intervention programme was carried out for eight weeks in Ability West, Mountbellew. Exercise classes took place twice per week during the clinical placement period of the student. While sixteen sessions were supposed to take place, the subject completed fourteen sessions due to the absence from two sessions not associated with the exercise programme. Exercise classes took between 30 to 40 minutes and involved a structured exercise session involving warm-up, Motomed exercise for upper and lower limbs, walking exercises, and cool down sessions. The subject showed high levels of adherence to the intervention programme and showed enthusiasm in each exercise class without needing much encouragement. The main element of the intervention comprised the use of Motomed movement therapy through active cycling. Upper limb exercises included approximately 14 minutes of active arm cycling, whereas lower limb exercises involved seven minutes of active cycling. The level of resistance used for the participant was adjusted depending on her ability and endurance and usually varied between Levels 3 and 4, sometimes progressing to Level 5 as her confidence increased and cycling capacity improved. It seemed that the participant tolerated well the increasing resistance and was able to perform continuous movements during most of the exercise sessions. Increasing of resistance was done depending on exercise tolerance rather than predefined increases of resistance.

After completing the Motomed program, the participant performed brief functional walks in the rehab setting. These exercises facilitated activation of lower limbs and development of confidence in mobility. Some functional exercises were also introduced, including movement exercises aimed at fostering coordination and independence in daily activities. Continuous supervision by the health care team was done throughout the intervention period. Special emphasis was put on observing the occurrence of symptoms such as tiredness, discomfort, or seizures. No seizures were observed in any of the exercise sessions conducted and the participant completed all the exercise sessions safely. The level of motivation and positive attitude of the participant to engage in exercise improved over the duration of the placement. As she became more familiar with the exercise program, she required less verbal encouragement and showed more confidence during the exercise sessions.

### 3.4 Outcomes of the Exercise Intervention

The participant successfully attended and participated in fourteen exercise sessions during the eight weeks of intervention at Ability West, Mountbellew. Despite an initial plan of sixteen exercise sessions, two sessions were missed due to being absent from the service. Nevertheless, overall participation in the intervention was high, and the participant displayed excellent motivation and enthusiasm throughout the exercise programme.

#### Functional Walking Capability

The participant's functional walking capability was measured prior to and after completion of the intervention using the Six-Minute Walk Test (6MWT). The participant successfully completed the test in accordance with her comfortable walking pace without needing extended rest periods. It could be stated that the walking capability was satisfactory in terms of her functional capability.

After the eight-week intervention, it was determined through observation that there was an improvement in walking confidence and exercise tolerance. Even though the ability of the participant to walk initially fell within an acceptable functional range, it seemed like he/she walked with more confidence after regularly engaging in the Motomed exercises and functional walking exercises. During the post-intervention assessment, the participant walked at a consistent pace for six minutes and seemed less tired than during the initial assessment. There was also noted an improvement in the endurance level of the participant during the routine rehabilitation activities, where he/she seemed to be willing to engage in physical activities for longer without asking for rest. These results indicate that regular participation in exercise might have led to improved functional endurance and confidence during walking exercises.

**Conclusion of Test** The test done at the end revealed that the patient had a constant walking rhythm in the six-minute walk test. The patient showed less fatigue after doing the test than when it was first conducted. Walking endurance is evident in the daily rehabilitation exercises because the patient has become more willing to exercise for longer durations without requiring any rest. This is an indication that consistent physical exercise has played a major role in the patient's performance.

### Grip Strength of Upper Limb

A reassessment of hand grip strength was conducted after the completion of the intervention program. Baseline grip strength was found to be clinically satisfactory in both hands. After fourteen sessions of the exercise regimen, there seemed to be an observed improvement in muscular function in the upper limb region. While significant improvements were not anticipated because of the short duration of the intervention program, repetitive upper limb cycling using the Motomed system seemed to have some positive effects on muscle endurance in addition to upper limb activation. The participant seemed to have better control over upper limb movements and could maintain sustained upper limb movement without any interruptions during subsequent exercise sessions.

## Exercise Endurance

Improvement in exercise endurance is one of the most apparent changes that occurred during the course of this study. In the beginning, the participant sometimes needed brief encouragement in order to sustain cycling cadence and finish the exercise period. Further on in the programme, her ability to perform upper and lower limb Motomed sessions without interruptions became better due to increased stamina. The exercise duration stayed the same through the whole programme, but resistance increased from Level 3-4 to occasionally Level 5 according to the participant's performance on the particular day. Nevertheless, she did the exercises with good cycling technique and no difficulty. It is also worth noting that improved endurance could be seen in the process of performing functional walking tasks after Motomed sessions. The participant had higher levels of energy, showed less signs of being tired and was more enthusiastic about exercising.

## Range of Motion

The visual inspection of joint movements did not show any changes in the range of motion of the upper and lower limbs during the intervention period. The absence of changes in range of motion was anticipated as the main goal of the intervention was the enhancement of functional endurance and exercise engagement. There was no loss in joint flexibility.

## Confidence and Exercise Involvement

Gradual increases in the confidence of the participant have been seen through the entire rehabilitation program. At the onset, verbal encouragement was needed in order to keep up the participant's motivation through exercise activities. As the participant got used to the Motomed program, she became more independent in her sessions and was more willing to get involved. She always came to the exercise sessions in good spirits and often expressed her pleasure in exercising with the use of the Motomed machine. She seemed to be becoming more enthusiastic in the course of the placement and became less prompted by the supervisory staff to start or carry out the exercise. Her growing confidence was also evident in functional walking activities, as well.

### Monitoring Seizures

Safety was the main concern during the intervention due to the fact that she had been diagnosed with mild tonic-clonic epilepsy. Seizures were not recorded during any of the sessions throughout the eight weeks. The subject did not experience any neurological problems while performing exercises with her arms and legs using the Motomed system. The participant did not experience any dizziness, extreme fatigue, unbalanced movements or any health-related problems during the exercise sessions. Absence of seizures during the entire intervention is consistent with previous research that states that exercising under supervision is safe for individuals with controlled epilepsy.

### Participant adherence

Adherence to the exercise regime was very good. The participant attended fourteen out of sixteen scheduled exercise sessions, achieving an adherence rate of around 87.5%. The missed sessions were due to absence from the service rather than any intolerance of the exercise or any medical difficulties. The participant cooperated well throughout all sessions and was highly engaged in the process.

### 3.5 Summary of Findings

Generally, the patient showed good response to the individualized exercise intervention conducted during the eight week placement despite the good baseline level of functional abilities. Improvements in exercise endurance, confidence when walking, upper limb function and participation in physical activities were noticed. The Motomed device proved to be well tolerated and offered a safe way to encourage repetitive movements of the upper and lower limbs in a controlled rehabilitation setting. Increases in cycling resistance were achieved progressively, demonstrating good physical tolerability of the intervention.

Possibly the most significant clinical finding is the fact that it increased the patient's enthusiasm, confidence and readiness to take part in exercises. This has serious implications for physical activity participation of adult people with intellectual disability since enjoyment and motivation are the recognized predictors of exercise adherence. There were no seizures or other exercise-related adverse events recorded throughout the intervention, which demonstrates that appropriate supervision of the exercise programs can be safely applied in patients with medically-controlled epilepsy. All these factors prove the potential of individualized Motomed-assisted exercises as a helpful intervention in the case of adults with intellectual disability and mild epilepsy.

## **Chapter 4: Discussion**

### **4.1 Introduction**

The aim of the present case study was to assess the impact of individualised exercise intervention on a 25-year-old woman diagnosed with intellectual disability and mild tonic-clonic seizures attending Ability West, Mountbellew, County Galway. The intervention involved predominantly supervised Motomed-assisted exercise of the upper and lower limbs along with functional walking for eight weeks. The programme focused on improvement of functional mobility, muscle endurance, exercise tolerance and confidence while guaranteeing the safety of the patient during the rehabilitation process.

Overall, the results showed positive clinical outcomes after the intervention. Changes occurred in the form of improvement in walking endurance, exercise tolerance, upper limb muscular performance and confidence during exercise. Notably, the subject participated in fourteen sessions of supervised exercise and no seizure attacks or adverse exercise related events occurred during the programme. Thus, the results of this case study are in agreement with existing data which states that exercise is a safe tool in the population of individuals with intellectual disabilities and controlled epilepsy (Shields et al., 2008; Arida et al., 2013).

### **4.2 Effect of Exercise on Functional Walking Ability**

One of the key findings of this study has been an improvement in the functional walking ability of the participant after the eight weeks of the intervention. Although the participant showed satisfactory walking ability in the baseline stage, improvement in endurance, confidence and movement continuity has been noticed after regular Motomed exercise and functional walking.

Walking ability is one of the significant criteria of functional independence among intellectually disabled adults. Low physical activity levels often lead to poor cardiovascular fitness, lower muscular endurance and limited community involvement. According to Shields et al. (2008), adults with intellectual disability usually have poor walking endurance as compared to those without disability due to sedentary lifestyle and lack of opportunity for physical activity (Hilgenkamp et al., 2012).

Findings from the current case study support the earlier findings that regular aerobic exercises improve functional mobility and walking endurance in adults with intellectual disability (Shields et al., 2008). Repeated movement of legs through cycling on the Motomed equipment is most likely to have resulted in better activation of leg muscles, higher muscular endurance and efficient cardiovascular function. It is likely that the functional tasks that were done after Motomed sessions have further ingrained the normal walking patterns as well, by virtue of the repetition principle. Studies have proven that exercises that include task-oriented movements along with aerobic training can help increase functional mobility in neurological rehabilitation (American College of Sports Medicine, 2022). Even though there were no measurements taken of improvement in walking distance since the baseline walking distance was within functional range, improvement could be seen from subjective observations.

#### 4.3 Upper Limb Function and Grip Strength

Upper limb training exercises comprised the bulk of the intervention program, with the participant performing about fourteen minutes of active arm cycling in each intervention session. After the intervention period, small improvements in grip strength and upper limb muscle endurance were noted. Grip strength is known to be an indicator of general muscle strength and functionality. Studies have shown that higher levels of grip strength are linked to greater independence during ADLs, improved functional performance and higher quality of life among neurological patients (Roberts et al., 2011).

The repeated nature of Motomed upper limb cycling exercises can account for the noted improvements. Arm cycling necessitates repeated contraction of the muscles in the shoulder, elbow and forearms, leading to muscle endurance and neuromuscular coordination. The fact that resistance was kept quite low during the whole program did not preclude functional improvements after fourteen supervised sessions.

Similar results have also been found through previous research conducted on arm cycling exercises, where upper limb endurance, as well as motor control and functionality, were enhanced by repetitive training (Werner et al., 2012). The current findings therefore lend weight to the inclusion of upper limb cycling as an intervention for adults with intellectual disabilities. What should be considered is that the period during which the intervention was carried out was rather brief. Hence, any significant changes in terms of muscle strength could



not expected. However, some improvement noted points to the potential positive effects of such activities.

#### 4.4 Exercise Endurance and Engagement in Physical Activity

The most clinically relevant outcome of the present case study intervention was improvement in the patient's endurance during exercise. Initially, at the start of the program, there were some instances where the participant appeared fatigued and needed verbal encouragement to participate in her exercise sessions. As the program progressed, however, the participant participated more regularly in her exercise sessions, tolerated higher Motomed resistance levels and took fewer breaks. There are several possible physiological adaptations that could explain the participant's improved exercise endurance, and these include cardiovascular efficiency, better oxygen utilization and increased muscle oxidative capacity (ACSM, 2022). Though these physiological factors have not been directly quantified, the ability of the participant to engage in exercises for longer periods may imply better exercise functionality.

Regular engagement in physical activity is highly beneficial for people with intellectual disabilities. These benefits include reduction in obesity, better cardiovascular fitness, improved psychological state and increased socialization (Temple et al., 2010). Promoting lifelong engagement in enjoyable physical activities should be a key objective for community rehabilitation programs like Ability West.

#### 4.5 Exercise Safety of Individuals with Epilepsy

The participant's medical condition, namely mild tonic-clonic epilepsy, was one of the major points taken into account during the whole process of the intervention. In the past, patients with epilepsy used to be discouraged from engaging in physical exercises due to the risk of developing seizures. Nevertheless, contemporary research suggests that physical activity is relatively safe when it is prescribed properly to patients who have epilepsy well under control (Arida et al., 2013). This case study confirms these suggestions. No seizures were reported to happen during any of the fourteen supervised physical activities involving upper and lower limb cycling and functional walking.

The success of the programme implementation could be attributed to several aspects. First of all, the intensity level of the exercise regimen was low and moderate; furthermore, the process

took place progressively and always in the safe environment. The supervision of all sessions was performed by the qualified personnel allowing the adjustment of physical activity depending on the patient's condition. Therefore, these results give more grounds for supporting physical activity among adults with both intellectual disability and epilepsy.

#### 4.6 Psychological and Behavioural Outcomes

Aside from the physiological changes witnessed in the participant, significant positive psychological and behavioural changes were also noted throughout the process. Increased self-confidence, enthusiasm and participation in exercise activities became some of the most promising results obtained in the current study. People with an intellectual disability tend to lack self-confidence when it comes to engaging in physical activities due to their communication problems, previous lack of exercise and care for the help of others. Therefore, behavioural changes can be as important as the physiological changes. As she was getting used to the programme, the participant seemed more confident when it came to using the Motomed device. The participant needed fewer prompts, participated more actively and even expressed her joy when doing exercises. These results coincide with previous research stating that a fun and individualized program increases motivation and improves adherence to the exercise activity. Self-confidence can affect socialization and quality of life. It is likely that the supportive atmosphere at Ability West as well as constant encouragement by the healthcare professionals helped in achieving these results.

#### 4.7 Comparison with Current Literature

Generally, the findings of the current case study are in agreement with the current literature on the advantages of the exercise interventions on individuals with intellectual disability and epilepsy. Previous research has consistently shown that physical exercise positively influences cardiovascular fitness, muscle endurance, functional mobility and quality of life among intellectually disabled adults (Shields et al., 2008; Hilgenkamp et al., 2012). Likewise, the current case study participant showed some improvements in walking endurance, exercise tolerance and self-confidence after taking part in the eight-week

intervention programme.

The use of upper limb cycling was particularly effective at increasing the participant's muscular endurance and motivation to exercise. As mentioned above, the repetitive actions associated with active arm cycling ensure continuous muscle work while limiting joint overload. It should be noted that this technique is especially appropriate for people with cognitive disabilities who are unable to complete traditional resistance exercises on their own.

Lower limb cycling with the use of a Motomed machine can also be considered the key factor in the development of walking endurance and overall body fitness of the participant. Repetitive movements of the legs contribute to the stimulation of activity of the quadriceps, hamstrings, glutes and calves while also enhancing cardiovascular endurance. Motomed-assisted exercise was the main type of intervention used in the current case study. Although little research exists in the literature on the use of Motomed in physically disabled adults, studies on neurological rehabilitation have shown positive effects on muscle activation, functional mobility and exercise tolerance after repetitive cycling intervention programmes (Werner et al., 2012). These findings show that the use of Motomed-assisted exercise is a safe and effective way of encouraging physical activity among intellectually disabled adults.

The behavioural enhancements observed during this case study have been backed by several past studies. A few pieces of literature have emphasised that enjoyment, socialisation, and individualised exercise programmes play a crucial role in determining the extent of physical activity participation among adults with intellectual disabilities. The individual participating in this study enjoyed all rehabilitation sessions and increased enthusiasm with the progression of the programme, indicating the importance of conducting enjoyable and person-centered exercises.

Moreover, the lack of seizures occurring throughout this intervention is also consistent with the current guidelines put forth by epilepsy organizations and the guidelines for exercise. According to the guidelines, physical activities that are supervised properly do not tend to increase the incidence rate of seizures among individuals with epilepsy who are on treatment. These results will serve as proof against the myths regarding exercise being forbidden in people with epilepsy.

Overall, these findings provide substantial evidence for the growing body of research that supports the importance of exercise physiology interventions as part of multidisciplinary rehabilitation of adults with intellectual disabilities and epilepsy.

#### 4.8 Clinical Implications for Exercise Physiology Practice

There are several significant implications of the current case study on the clinical practice of exercise physiology. Firstly, the current results show that adults with an intellectual disability and medically controlled epilepsy may take part in a structured program of exercises provided that the program is individualized taking into account their health and functioning condition.

The application of Motomed-assisted exercises is associated with several advantages in the rehabilitation process carried out in a community setting. Due to the fact that the exercise takes place in a sitting position, the risks of falls are significantly lowered. Besides, the adjustment of the level of resistance allows adjusting the intensity of exercise to individual possibilities and performing repetitive exercises of the upper and lower body simultaneously. Finally, one more important implication is related to the importance of supervision. The continuous supervision was helpful not only in making participants feel safe but also allowed modifying the exercise regimen in case of necessity.

It is important for exercise physiologists when dealing with intellectually disabled patients to take into consideration that behavioural changes may have the same significance as physiological ones. Greater confidence, joy and motivation to engage in exercising activities will most certainly have an effect on adherence to exercise and might lead to improved lifestyle. Lastly, the current case study has demonstrated the need to apply a person-centered approach to the rehabilitation process. Focusing not only on impairments but rather on abilities, likes and goals of a participant of exercise programs will help achieve the desired results.

#### 4.9 Advantages of the Case Study

Some advantages need to be taken into account during the interpretation of the research results. To begin with, the intervention took place in a natural environment of community rehabilitation. In other words, the findings become more applicable to routine clinical practice since they were obtained not in artificial laboratory settings. Another advantage was that the participant attended most of the planned exercise sessions and showed good compliance during all of them. It was more likely that changes seen by the researcher were caused by the effect of the exercise program. Individualization of the intervention is another advantage. The exercise prescription was adjusted according to physical capacities, exercise tolerance, and performance of the participant on a daily basis. Supervision during all exercise sessions became an additional benefit due to the presence of tonic-clonic epilepsy in the participant. The fact that there were no seizures or other adverse effects makes it feasible to develop supervised exercise programs.

Ultimately, the combination of objective testing and clinical observation led to a thorough assessment of participant development. Though objective data contributed to an understanding of participants' functional performance, the observational data provided an important insight into their confidence and motivation.

#### 4.10 Limitations of the Case Study

However, some limitations need to be addressed despite the positive results obtained in this research. First, the biggest limitation is that the sample of the study consisted of only one individual. Therefore, the results cannot be generalized to all people with intellectual disability and epilepsy, since different levels of cognitive ability, physical performance, health status and exercise tolerance exist.

The second limitation is related to the relatively short period of time during which the intervention took place, which lasted for eight weeks only. It is expected that even bigger physiological changes may take place during longer intervention. Thirdly, the choice of outcome measures applied in this research needs improvement. Even though the use of the Six-Minute Walk Test and Grip Strength Test gave some valuable information about the level of functional performance, the addition of balance test, gait analysis and quality of life questionnaire could provide a more holistic picture. However, the balance test was deemed inappropriate due to the limitations of cognitive ability of the participant.

Progression of the exercise program was dependent more on clinical observation than on physiological markers. While this is typical of clinical practice in community-based rehabilitation, it might result in reduced accuracy in prescribing exercises in comparison to laboratory-based rehabilitation studies. However, despite these constraints, the current case study offers relevant practical information that can be used to justify the use of supervised exercise programs in community services for people with intellectual disabilities and epilepsy.

#### 4.11 Recommendations for Future Research

Further research should address the effectiveness of Motomed-assisted exercise on greater samples of adults with intellectual disability and epilepsy. Randomised controlled trials of Motomed exercise compared with conventional exercise programs could provide clearer evidence concerning the efficacy of such an intervention program. In addition, studies should investigate the use of longer intervention periods which extend beyond eight weeks to find out whether there is any development in muscular strength, cardiovascular capacity and functional independence after those periods.

Moreover, further studies should include the use of other outcome measures such as balance test, gait assessment, quality of life scales, physical activity measures and comments from the caregivers. The study of psychological effects of participation in exercise by people with intellectual disability could also be interesting. Confidence, motivation and increased socialisation are important outcomes of an intervention which might affect future health-related behaviors but have not been sufficiently represented in the literature until now. Finally, future research should focus on a multidisciplinary approach to rehabilitation through exercise physiologists, physiotherapists, occupational therapists and nurses.

## 4.12 Chapter Summary

In this chapter, the findings of this case study have been reviewed in light of existing literature on intellectual disabilities, epilepsy and exercise rehabilitation. In general, the participant achieved improvements in terms of walking endurance, upper-limb mobility, exercise tolerance and confidence after completing an eight-week programme of supervised Motomed exercise. Most importantly, there were no seizures during this intervention, which shows the safety of exercise therapy in cases where the epilepsy is well-controlled.

A comparison of the results obtained in this case study and the results of previous studies revealed that our findings are in line with evidence on the positive effects of physical activity in people with intellectual disabilities. This is further emphasized by the increased enthusiasm and confidence in the participant, which makes person-centred rehabilitation programmes that promote physical activity crucial. Even though the limitations of sample size and intervention period should not be overlooked, this case study gives useful clinical information about the use of Motomed exercise in a rehabilitation setting. It can be used as a basis for future research on the topic of exercise programmes for adults with intellectual disabilities and epilepsy.



## **Chapter 5: Conclusions and Recommendations**

### **5.1 Conclusions**

The current case study involved evaluating the effectiveness of an individualized exercise intervention for a 25-year-old female patient with an intellectual disability since birth and mild tonic-clonic epilepsy who is participating at Ability West, Mountbellew, County Galway. The intervention was performed for eight weeks during an Exercise Physiology clinical placement and involved supervised exercise of the limbs using Motomed machine and functional walk.

Adults with intellectual disabilities often suffer from limited physical activity, low cardiovascular fitness, poor muscular strength and limited access to an exercise program. In addition, their condition is often accompanied by epilepsy, which also reduces physical activity due to concerns about seizures, safety and confidence. Thus, developing safe, individualized and enjoyable exercise intervention is a vital part of the rehabilitation process.

The results of this case study have shown that the individual has responded to the intervention program very positively. Despite the fact that the initial level of functionality had already been quite good, some changes have been made in terms of increased walking endurance, increased tolerance to physical activity, improvement of the upper extremity function and confidence in physical activities. During the course of the intervention the participant has been showing increased eagerness to perform physical exercises, improved performance during exercising, as well as greater enjoyment of physical activity. These behavioral changes are considered clinically significant since the confidence and enjoyment are known to be factors which influence the adherence to the exercise program. The use of the Motomed movement therapy device has turned out to be very effective and convenient for the rehabilitation process of this participant. The active cycling of the upper extremities for about fourteen minutes and the active cycling of the lower extremities for about seven minutes per session had been providing repetitive exercise patterns with safety for the participant.

Another interesting result of the current research is the lack of seizures during the whole 14 exercise sessions conducted. No neurological disorders occurred in the process of performing the exercise programme by the participant, which proves the safe implementation of the properly supervised physical activity in patients with medically managed mild tonic-clonic epilepsy. This result is also consistent with modern evidence suggesting the necessity of .

Conducting physical activities in people suffering from epilepsy provided that safety measures are performed correctly. In addition, the achieved results regarding functional walking ability are also of clinical significance. Despite the fact that the participant successfully completed the 6-minute walk test, he experienced increased confidence and endurance after completing the exercise programme. In the same way, minor changes in grip strength and muscle endurance of upper limb muscles were reached after conducting repetitive active upper limb Motomed exercise. However, no changes in range of motion were observed during the process of the programme implementation due to the patient's neurological status and the short intervention period.

It was once again highlighted through this case study that it is essential to follow a person-centred approach when it comes to developing exercise prescriptions. In this case, the exercise program has been developed on the basis of the physical capacities of the participant as well as his motivation and performance in day-to-day life instead of going strictly by any exercise protocol. The continuous monitoring of the process by health professionals ensured safety and helped in developing independent and participative approaches. Although this case study includes just one participant and cannot be generalized to all people, it offers much insight into how exercise physiology can be used in community-based rehabilitation programs. The findings of this study have proven the need for including the exercise programs in the treatment plans of adults with intellectual disability and epilepsy. There is a major role for exercise physiologists when it comes to enhancing physical health, functional capabilities, and active physical participation among developmentally disabled persons.

Overall, the current study was successful in accomplishing its main objective of assessing the efficacy of the implementation of a program of individualized exercises. It revealed that the program had a positive impact on participants' functional endurance, self-confidence, and their participation in exercise programs without any associated risks to their health and wellbeing. The results add to existing evidence regarding the safety and effectiveness of exercise as therapy in intellectual disability and mild epilepsy.

## 5.2 Clinical Recommendations

From the results obtained from the current case study, the following recommendations can be proposed. Exercise regimens should be individualised depending on the individual's functionality level, history of medical problems and personal preferences. Personalised interventions have the potential to increase motivation levels, exercise adherence and participation. Health practitioners are advised to recommend regular participation in planned physical activities as opposed to not exercising due to the person's intellectual disability and well-managed epilepsy. Safety precautions, including appropriate supervision, can make it possible for the individual to exercise safely and reap health benefits.

The use of Motomed to perform exercise regimens should be used as a good rehabilitation tool among community disability services because of the design of the device. Its capability of accommodating seated exercises and variable levels of resistance as well as the active-assisted movements makes it more appropriate for people with such disorders. Exercise physiologists should cooperate with physiotherapists, occupational therapists, nurses, support workers and relatives of the client to design a comprehensive programme.

Functional evaluations should form an integral part of the rehabilitation programs in order to assess participants' progress and plan exercises appropriately. Outcome measures such as gait performance, strength, balance and quality of life can help clinicians assess the effectiveness of interventions. Engagement in a physically active lifestyle should always be the main focus of rehabilitation programs. Exercise performed outside the confines of the clinic may lead to better health outcomes and improved quality of life.

### 5.3 Recommendations for Further Studies

Further studies with bigger samples are needed in order to provide evidence for the effectiveness of Motomed-assisted exercise in adults who have intellectual disability and epilepsy. The randomised controlled trial where Motomed treatment will be compared to conventional exercises will allow obtaining additional information about the effectiveness of interventions. In future studies, it is necessary to study the impact of exercise interventions not only up to eight weeks. Prolonged interventions can lead to greater results in terms of cardiovascular fitness, muscle strength, body composition, and independence in activity. Researchers should use a broader list of variables as the indicators of exercise effectiveness, including balance testing, gait analysis, quality of life questionnaires, fatigue scales and subjective outcomes provided by carers.

Further research on the effects of various levels of exercise intensity, frequency and technology may aid in the creation of evidence-based recommendations of exercise programs for people with intellectual disabilities and epilepsy. Research into exercise programs conducted at home as well as digital technology for rehabilitation may also help make physical activity more accessible for people residing in the community.

Lastly, research in the future will need to focus on the use of multidisciplinary rehabilitation for adults with intellectual disabilities. The collaboration of exercise physiologists, physiotherapists, occupational therapists, psychologists and other medical practitioners may help develop a more holistic approach to rehabilitation programs.

## 5.4 Concluding Statement

In summary, it is evident from this case study that a properly structured exercise program including both upper and lower limbs Motomed-assisted exercise could be conducted safely and effectively on a patient with intellectual disability and mild tonic clonic seizures. The benefits of the intervention included increased exercise tolerance, functional ambulation abilities, upper limb function, and increased confidence, with no occurrences of seizures or any other adverse events during the course of rehabilitation.

While more studies need to be conducted in order to reinforce current knowledge, the results of the case study provide support to the increasing understanding of the significance of exercise as an important part of the rehabilitation process for people with intellectual disability and epilepsy. Exercise physiologists make an invaluable contribution to this end.

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